

Topology First-Year Exam, May 2025, Part 2

For the first five problems, show all your work and support all statements.

1. Let $p : E \rightarrow B$ be a covering map, and let $p(e_0) = b_0$. Define the map $\Phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0)$ which gives the lifting correspondence. Prove that Φ is surjective if E is path connected, and bijective if E is simply connected.
2. Prove that $\pi_1(S^1, 1) \cong \mathbb{Z}$.
3. This problem has two parts.
 - (a) Define the Hawaiian earring space, H .
 - (b) Show the images of the generators of the fundamental groups of the circles do not generate the fundamental group of H .
4. This problem has two parts.
 - (a) Define surface.
 - (b) Compute the fundamental group of the connected sum of a torus and second torus $T^2 \# T^2$.
5. Let X be the quotient space obtained from an 8-sided polygonal region P by pasting its edges together according to the labeling scheme $abcdc^{-1}a^{-1}db$. It turns out that all vertices of P are mapped to the same point of the quotient space X by the pasting map. Calculate $H_1(X)$, and using this, determine which compact surface X is homeomorphic to.

Answer the following with complete definitions or statements or short proofs.

6. State the Tietze Extension Theorem.
7. Show that if $h, h' : X \rightarrow Y$ are homotopic and $k, k' : Y \rightarrow Z$ are homotopic, then $k \circ h$ and $k' \circ h'$ are homotopic.
8. Let E and B be topological spaces. Define what it means for a function $p : E \rightarrow B$ to be a covering map.
9. If $j : A \hookrightarrow X$ is the inclusion of a retract, then show that the induced map on fundamental groups is injective.
10. Let α be a path in X from x_0 to x_1 . Define the map $\hat{\alpha} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ and show that $\hat{\alpha}$ is a group isomorphism.
11. State the Seifert–van Kampen Theorem.
12. Let S^n be the n -sphere. Prove that $\pi_1(S^n)$ is trivial for $n \geq 2$.
13. State the Borsuk–Ulam Theorem for S^2 .
14. State the Brouwer Fixed Point Theorem for the disk.
15. Give an example of a topological space whose fundamental group is $\mathbb{Z}/3\mathbb{Z}$. Give an example of a topological space whose fundamental group is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.